

YOUNGHO KIM

Curriculum Vitae

Personal Information

AFFILIATION Korea Advanced Institute of Science and Technology (KAIST)
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ADVISOR Kuk-Jin Yoon
BIRTH / NATIONALITY February 20, 2001 / Republic of Korea
LANGUAGE Native Korean, fluent in English

Education

CURRENT	Korea Advanced Institute of Science and Technology M.S. in Mechanical Engineering (GPA: $\sim$ 4.3) Advisor: Kuk-Jin Yoon Thesis: –	Daejeon, Korea
AUG. 2025	Korea Advanced Institute of Science and Technology B.S. in Mechanical Engineering (GPA: 4.11/4.3), Summa Cum Laude	Daejeon, Korea
FEB. 2019	Hana Academy Seoul	Seoul, Korea

Research Interests

I am interested in **robotics, computer vision, and embodied AI**. This includes broad topics of vision-language-action models, robot learning, event-based perception, pose estimation, and robust visual representation learning for real-world robotic systems. Recently, I have been particularly working on the following topics:

- Event-based Vision-language-action models for dynamic manipulation
- Visual representations learning for monocular VLA
- Event-based Human pose estimation
- Reinforcement learning and sequential decision-making for perception systems

Publications

*Equal contribution.

Under Review

- [1] GoStop: Reinforcement Learning for Adaptive Temporal Aggregation. *Under review*.

Peer-Reviewed Publications

- [2] Hoonhee Cho*, Jae-Young Kang*, Yuhwan Jeong*, Yunseo Yang, Wonyoung Lee, **Youngho Kim**, and Kuk-Jin Yoon. “DSERT-RoLL: Robust Multi-Modal Perception for Diverse Driving Conditions with Stereo Event-RGB-Thermal Cameras, 4D Radar, and Dual-LiDAR.” *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2026.

- [3] Jae-Young Kang*, Hoonhee Cho*, Taeyeop Lee*, Minjun Kang, Bowen Wen, **Youngho Kim**, and Kuk-Jin Yoon. “Event6D: Event-based Novel Object 6D Pose Tracking.” *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2026.
- [4] Taeyeop Lee*, Gyuree Kang*, Bowen Wen, **Youngho Kim**, Seunghyeok Back, Inso Kweon, David Hyunchul Shim, and Kuk-Jin Yoon. “DeLTa: Demonstration and Language-Guided Novel Transparent Object Manipulation.” *arXiv preprint*, 2025.
- [5] **Youngho Kim***, Hoonhee Cho*, and Kuk-Jin Yoon. “From Sharp to Blur: Unsupervised Domain Adaptation for 2D Human Pose Estimation Under Extreme Motion Blur Using Event Cameras.” *IEEE/CVF International Conference on Computer Vision (ICCV)*, 2025.
- [6] Hoonhee Cho*, Jae-Young Kang*, **Youngho Kim**, and Kuk-Jin Yoon. “Ev-3DOD: Pushing the Temporal Boundaries of 3D Object Detection with Event Cameras.” *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025. **Highlight, top 3.0%**.

Research Experience

Undergraduate Researcher, KAIST

August 2024 – June 2025

VILAB, advised by Kuk-Jin Yoon

- Participated in research on Event-based 3D object detection. Constructed the training dataset where RGB images, lidar and events are aligned. Presented the algorithm which is accurate and has low-latency.
- Conducted research on unsupervised domain adaptation. Presented machine learning pipeline that effectively reduced domain gap between blurry image and sharp image via event modality.

Undergraduate Researcher, KAIST

September 2023 – July 2024

RAILAB, advised by Jemin Hwangbo

- Participated in individual project that learning synergies between pushing and grasping for manipulation, and a unified policy for manipulation and locomotion.

Awards & Honors

FALL 2024	KAIST ME, Dean’s List Award <i>Awarded to the top 3% of all ME students</i>
FALL 2023	KAIST ME, Academic Excellence Scholarship <i>Awarded to the top 4 students in ME</i>
FALL 2023	KAIST ME, Dean’s List Award <i>Awarded to the top 3% of all ME students</i>
FALL 2019	KAIST, Freshman Dean’s List Award <i>Awarded to the top 3% of all freshman students</i>

Skills

Programming Tools. Python, PyTorch, MATLAB, ROS, OpenCV

Extracurricular Activities

2023 – 2024	KAIST KAKI (Snowboard Club) President
2021 – 2022	Korea ARMY Member
2019 – 2020	KAIST CHORUS (Chorus) Member